SEAC□□□

“ ”

“ ” “ ”

SEAC
Structure- and Evidence-Aware Chunking

1 –

2 –

3 10 89
66981 125

4 0.05

2 3
4 SEAC 5 6
7 8 9
10

2 相关工作

2.1 检索粒度与上下文保持

[2] Dense X Retrieval
[5] Late
Chunking
[6] RAPTOR [7]
SEAC

SEAC

2.2 结构化技术文档

Docling [8] “
” “ ”

2.3 软件工程追溯与检索治理

[9-10]
– –
“ ”
LangChain
LlamaIndex [11-12]

2.4 研究空缺与本文边界

SEAC “ ” OCR

MRR
MRR

“ ”

3 问题定义

3.1 可追溯单元与父子块

$$D = (u_1, u_2, \ldots, u_n).$$

u_i

h r_j

$$t(r_j) = \text{header}(h) \Vert \text{row}(r_j).$$

3.2 来源正确性与文本支持

q G(q) z(q) e_i

G(q) 1

G(q) “ ” z(q)

“ ” k I_h I_path I_group

3.3 研究问题

RQ1 SEAC “ ” 7

RQ2 SEAC RQ3

RQ4

RQ5

4 SEAC 方法

4.1 单元规范化与受保护结构

SEAC

`document_id` `block_id`

`unit_id` “

”

4.2 相邻边界评分

(u_i,u_{i+1}) d_i s_i

k_i p_i BGE-M3

k_i 10 1

V tokens(u)

“ ”“ ”“ / ” “ / ”

UTF-8

4.3 父块线性组装

512 BGE

320 0.03 224 48

b_i

4.4 细粒度子块与父块返回

112 176

“ ” SEAC

p q S_p S_c

工业文档 标题·段落·表格	→	可追溯单元 块/单元/标题路径	→	父块与子块 结构保护·行级索引	→	稠密检索 子块激活父块	→	证据输出 上下文+精确来源

1 SEAC

4.5 复杂度与可复现接口

n SEAC n n-1
O(n)

SHA-256 Python

4.6 合成示例与失败判定

“ →
→TASK_ITEM” “ ”
“STATUS VARCHAR 16 ”
“STATUS VARCHAR 16 ” “
” “ ” STATUS
SEAC “ → →TASK_ITEM
STATUS VARCHAR
16 ” `unit_id`

`trigger_unit_id` `parent_chunk_id`
`source_block_id` “ ”

G(q) G(q)

STATUS
SEAC

“TASK_ITEM” –
512

5 工业实验设计

5.1 语料、项目隔离与文档类型

10 P01—P10 89

DOCX XLSX 6
PDF PPTX

P01—P02 P03
P04—P10

1

1

划分	项目数	文档数	解析块数	单元数	正则词元数
	2	15	9639	21371	418541
	1	7	3075	5193	103562
	7	67	19127	40417	1155797
	10	89	31841	66981	1677900

125 125

5.2 边界参考构造

P01 4783 P02—P10
300 SEAC
100
Cohen κ
0.70 [13]

κ
P08 κ=0.650
5566 77.5% 387
5.4% 1230 17.1% F1

20260804

κ

/ κ P02 0.61/0.87 P03 0.56/0.76 P04
0.81/0.89 P05 0.66/0.96 P06 0.42/0.91 P07
0.75/0.98 P08 0.68/0.65 P09 0.75/0.72 P10
0.77/0.76 κ κ
P08 “ ”

`HUMAN_VERIFIED`
`RULE_4_1`
`LLM_PRELABEL`
NOISE F1

5.3 问题与证据参考

215 195
10
P10 11 15
P04—P09 125

P04—P07 P09 P10 1368 2

9 195

`NO_RISK` `ANONYMIZED`
“ ” `CLEARED`
262 71 124 67

12
1—8 18
12

2

项目	文档数	单元数	边界评价	检索问题数
P04	17	8532		22
P05	17	8166		17
P06	12	10122		22
P07	8	2496		23
P08	5	4369	κ=0.650	22
P09	4	5324		19
P10	4	1408		11<15

5.4 比较方法与公平控制

BGE-M3 512
1 2 3
4 5 LangChain RecursiveCharacterTextSplitter 1.1.2 6 LlamaIndex SentenceSplitter 0.14.23 7 SEAC
8 SEAC [16]

LumberChunker
[17-19] BERGEN [20] SEAC

SEAC

5.5 指标、推断与实验环境

NOISE F1
MRR Hit@1/3/5 MRR Hit@1/3/5 MRR Hit@5

10000 [14]
SEAC LlamaIndex MRR 6 64
Holm [15]

“ ” Hit@k [21]

Python 3.12.3 `sentence-transformers` 5.6.0 `langchain-text-splitters` 1.1.2 `llama-index-core` 0.14.23 Linux ARM64 NVIDIA CUDA
16 20260804 3 SEAC
P01—P10

3 SEAC

	取值	作用
	BGE-M3	
	512/320/224/48	
	112/176	
	0.45 0.30 0.25	
	0.8 +0.08 −0.05	
	0.35/0.65	
	10000	
	Python 3.12.3 Linux ARM64 CUDA	

5.6 泄漏控制和完整性门槛

P04
—P10 QA

`industrial-seac-formal-analysis-v1.1`

5.7 可审查性与不可公开数据下的复现层级

“ ” “ ”

“ ”

“ ”

6 实验结果

6.1 总体边界与检索质量

4 SEAC F1 0.5673 95% [0.4708, 0.6488] SEAC 0.5032 SEAC 6 Holm p=0.4375 F1 0 125 SEAC MRR 0.7937 95% [0.7060, 0.8772] MRR 0.8561 Hit@5 0.9520 SEAC MRR 0.7290 0.0647 LlamaIndex MRR +0.1003 95% [0.0042, 0.2008] p=0.0625 0 0.05 6 64

4

方法	边界 **F1**	单元 **MRR**	**Hit@1**	**Hit@5**	片段 **MRR**
	0.0000	0.5158	0.3680	0.7200	0.1350
	0.0369	0.6086	0.4480	0.8080	0.7072
	0.5032	0.7290	0.6080	0.8880	0.8211
	0.3843	0.6999	0.5680	0.8640	0.8121
LangChain	0.0054	0.6460	0.4960	0.8480	0.7329
LlamaIndex	0.0920	0.6934	0.5520	0.8880	0.7586
SEAC	0.5673	0.6796	0.5280	0.8800	0.7770
SEAC	0.5673	0.7937	0.6800	0.9520	0.8561

6.2 跨项目一致性

SEAC

“SEAC

”

5 SEAC MRR SEAC P04 P05 P07 P08 P09 P06 P05 P06 0.0402 P06

5 SEAC LlamaIndex

项目	问题数	**SEAC MRR**	**LlamaIndex MRR**	差值
P04	22	0.6914	0.5033	+0.1881
P05	17	0.8627	0.6275	+0.2353
P06	22	0.6805	0.7206	−0.0402
P07	23	0.8080	0.7360	+0.0719
P08	22	0.8386	0.7224	+0.1162
P09	19	0.9123	0.8557	+0.0566

6.3 不同问题类型

5

MRR Hit@5 1.0000 MRR 0.9839 MRR MRR 0.5429 Hit@5 0.8000 0.7323 0.2516

6

6 SEAC

问题类型	数量	单元 **MRR**	**Hit@5**	片段 **MRR**
	17	0.8211	0.9412	0.8211
	10	0.7833	1.0000	0.7833
	16	0.9167	0.9375	0.9167
	31	0.7323	1.0000	0.9839
	23	0.7867	0.9130	0.7867
	9	0.8370	1.0000	0.8370
	9	1.0000	1.0000	1.0000
	10	0.5429	0.8000	0.5429

6.4 组件消融

0.8640 MRR 0.8289

7 SEAC 0.8129 Hit@5 0.9280 MRR

MRR 0.7937 0.6853 Hit@5 0.9520 0.0042

7

变体	边界 **F1**	单元 **MRR**	**Hit@5**	相对完整 **MRR** 变化
SEAC	0.5673	0.7937	0.9520	—
	0.5698	0.7937	0.9520	+0.0000
	0.5639	0.7959	0.9520	+0.0021
	0.5718	0.7979	0.9520	+0.0042
	0.5731	0.7959	0.9520	+0.0022
	0.5673	0.8129	0.9280	+0.0192
	0.5673	0.6853	0.8640	−0.1084
	0.5673	0.8289	0.9520	+0.0352

SEAC “ ” Holm p=1.0000 0.1084

6.5 资源开销

6 8891 SEAC 48912 5.50 SEAC 15.18 ms 20.93 ms 5.74 ms 8 SEAC

8

方法	向量数	分块时间/**s**	索引时间/**s**	查询均值/**95** 分位/**ms**
	8891	0.25	139.46	15.18/18.40
SEAC	48912	112.11	351.12	20.93/27.28

0.9520 4.8%

MRR MRR

6.6 排名结果的工程解释

“ ” “

MRR 0.6934 0.7937 ”

5 1 20

10 Hit@5 SEAC Hit@5

F1 MRR
SEAC -only 0.5673 F1
MRR F1

P06
P06 SEAC
5.50

“ ”

SEAC 0.8289

7 面向智能化工业软件工程的应用与治理

7.1 大模型驱动的需求分析

SEAC
“ ”
“ ” “
”

`document_id/block_id/unit_id`

SEAC “ — —
”

—

7.2 质量保障与幻觉审计

MRR MRR

“ ” —

SEAC

7.3 版本、访问控制与责任追踪

SEAC

7.4 工程集成边界

SEAC 5.50

MRR MRR

“ ”

7.5 软件生命周期中的落点

“ — — — — ”

SEAC

SEAC

8 讨论

8.1 当前证据支持的机制

“ ” SEAC
MRR 0.6796 0.7937 0.6853

MRR

8.2 父子融合与标题上下文的反向结果

0.8289 MRR

MRR Hit@5
“ ”

SEAC SEAC
0.8289

8.3 下一轮评价协议

6

9 有效性、伦理与数据可用性

9.1 构念与内部有效性

F1
77.5%
SEAC P08

66981
F1

9.2 结论与外部有效性

6
64

p=0.0625

PDF 6
PDF/PPTX
—

9.3 隐私、公开与人工责任

10 结论

SEAC

10 SEAC 125
MRR 0.7937 Hit@5 0.9520 SEAC
LlamaIndex p=0.0625

“ ” “ ” MRR
0.6853 0.8289
SEAC 5.50

SEAC

附录 A 父块与子块构造过程

— —

1
“
+ ”
`unit_id` `document_id` `block_id`

2 512

3

d_i s_i k_i p_i b_i

4 0.8

0.03

5

– JSON 0.7937

2 0.9520 0.7290 0.1003 [0.0042,0.2008]

p=0.0625 0.6853 0.8289 5.50

6

112 176 PDF Word

7

10 0

G(q)

z(q)

8

MRR

附录 B 复现、审稿与部署检查项

/

附录 C 项目与补充指标明细

SHA-256 Python A1 10 BGE

WordPiece

P08 κ=0.650 P10 11<15 P04—P10

A1

项目	划分	文档	解析块	单元	近似词元	表格	表格行
P01		7	4790	10574	194119	839	6623
P02		8	4849	10797	224422	986	6934
P03		7	3075	5193	103562	412	2530
P04		17	4180	8532	301754	474	4826
P05		17	3886	8166	228920	464	4744
P06		12	5623	10122	255278	758	5257
P07		8	1564	2496	59925	172	1104
P08		5	1206	4369	123307	198	3361
P09		4	1606	5324	133710	329	4047
P10		4	1062	1408	52903	62	408
	–	89	31841	66981	1677900	4694	39834

A2 κ≥0.70

P08

A2

项目	修订前 κ	修订后 κ	原始一致率/%	边界资格
P02	0.61	0.87	94.0	
P03	0.56	0.76	88.8	
P04	0.81	0.89	96.3	
P05	0.66	0.96	98.8	
P06	0.42	0.91	97.2	
P07	0.75	0.98	98.8	
P08	0.68	0.65	84.5	
P09	0.75	0.72	87.6	
P10	0.77	0.76	89.9	

A3 4 SEAC -
only

A3

方法	精确率	召回率	**F1**	**F1** 的 **95%** 置信区间
	0.0000	0.0000	0.0000	[0.0000, 0.0000]
	0.0402	0.0340	0.0369	[0.0051, 0.0783]
	0.3611	0.8298	0.5032	[0.4108, 0.5859]
	0.2920	0.5617	0.3843	[0.2779, 0.4800]
LangChain	0.0074	0.0043	0.0054	[0.0000, 0.0265]
LlamaIndex	0.4615	0.0511	0.0920	[0.0367, 0.1570]
SEAC	0.4321	0.8255	0.5673	[0.4704, 0.6486]
SEAC	0.4321	0.8255	0.5673	[0.4708, 0.6488]

A4 Hit@3 Hit@5 SEAC
Hit@3 0.9040 5 Hit@5
Hit@5 0.9520

A4

方法	单元 **MRR**	**Hit@1**	**Hit@3**	**Hit@5**	片段 **MRR**	片段 **Hit@5**
	0.5158	0.3680	0.6240	0.7200	0.1350	0.1840
	0.6086	0.4480	0.7200	0.8080	0.7072	0.8480
	0.7290	0.6080	0.8320	0.8880	0.8211	0.9040
	0.6999	0.5680	0.8240	0.8640	0.8121	0.8960
LangChain	0.6460	0.4960	0.7440	0.8480	0.7329	0.8720
LlamaIndex	0.6934	0.5520	0.8400	0.8880	0.7586	0.8960
SEAC	0.6796	0.5280	0.8160	0.8800	0.7770	0.8960
SEAC	0.7937	0.6800	0.9040	0.9520	0.8561	0.9520

致谢与人工智能使用声明

OpenAI Codex

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Background

Large language models are increasingly embedded in software engineering workflows, but industrial software differs from open-domain text generation in the density of constraints, the heterogeneity of artifacts, and the consequences of unsupported outputs. Requirements, interface specifications, database dictionaries, detailed designs, technical agreements, and meeting records jointly define system behavior. The same term may refer to different modules or versions, while critical evidence is often expressed in a table row whose meaning depends on its header and section path. A retrieval-augmented generation system therefore needs more than semantically similar passages: it needs evidence that remains attributable to the correct artifact location.

Document chunking determines the units visible to a retriever. Fixed windows can cut through a requirement or table. Generic recursive splitters follow separators but do not understand engineering coupling. Heading-only splitters preserve sections yet may return excessive context for field-level questions. Semantic splitters may identify topical shifts while separating a data row from its schema. These errors propagate to downstream generation, citation, consistency checking, and change-impact analysis. Long-context models do not remove the problem because irrelevant context increases cost and evidence can still be overlooked or confused.

SEAC addresses this representation layer. It normalizes paragraphs, headings, and table rows into stable source units, constructs coherent parent chunks with deterministic boundary rules, and indexes fine-grained children that retain heading paths and table headers. A child is used for matching, while its parent provides interpretation context. The retrieval record preserves the exact document, block, and unit identifiers that produced the result. This separation enables two complementary evaluations: whether returned text contains the supporting span and whether it originates from an accountable source unit.

The industrial study follows a frozen protocol because project templates and terminology create clustered observations. Development, validation, and

locked-test projects are separated. Questions and boundary references are reviewed before locked metrics are released. Statistical inference resamples or exchanges labels at the project level rather than treating all questions as independent. The study reports negative results and resource costs alongside aggregate gains. In particular, the available evidence supports header-aware table-row retrieval, but it does not validate every component of the full scoring design or establish general superiority with only six independent retrieval projects.

For intelligent industrial software engineering, provenance-aware retrieval can support requirements extraction, interface and database consistency checking, test generation, defect review, and audit logs. It must nevertheless be combined with document-level access control, versioned parent-child mappings, privacy review, and human responsibility. Retrieved evidence can improve the verifiability of a large-model workflow, but it does not by itself authorize automated decisions or guarantee that a generator will faithfully use the evidence.